

multiVector.net

A live, expression-driven
Geometric Algebra visualization environment

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Why this matters

The GA tooling gap

Geometric Algebra is winning the theoretical battle in graphics, robotics and physics. But the tooling story has not kept pace.

- `ganja.js` — powerful but bare; every demo is hand-rolled.
- `GAlgebra` / `Clifford.jl` — symbolic, no live interaction.
- `CGA viewer` — single algebra, fixed UI.

No project today gives the user:

- a uniform expression language across algebras,
- an interactive canvas where every object is draggable, and
- a clean way to define their own primitives.

The opportunity

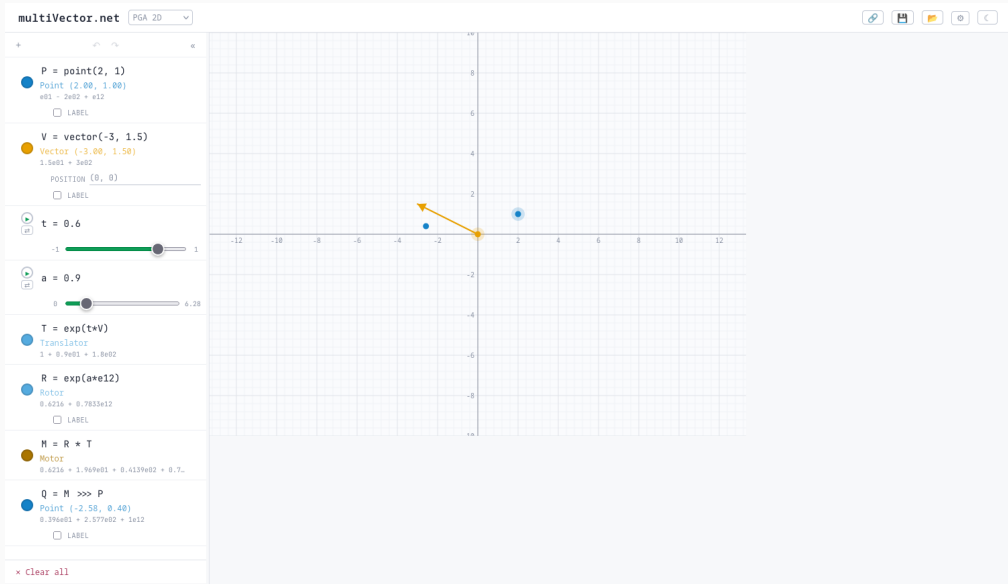
GA needs a **Desmos-style standard** for visualizing multivector constructions — but mathematically faithful, algebra-pluggable, and open.

multiVector.net aims to be that standard.

- One language, many algebras.
- Expressions *are* the construction — drag a point, every dependent object updates.
- User-extensible: define your own primitives in the language itself.

What it is

In one screenshot



The expression language

Every line is parsed, classified, evaluated, and rendered.

```
P = point(1, 1)           # free draggable point
Q = point(2, 2)
L = P & Q                 # join: line through P and Q
M = exp(0.5 * e12)        # rotor (motor)
Q' = M >>> P             # sandwich transform
t = 0.5                   # animatable scalar slider
T = exp(t * (e01 + e02))  # translator parameterised by t
```

Operators (tight $\rightarrow$ loose): `! ~ > ^ & | § > * / > >>> > + -`

Type is a property of value, not syntax

Any expression is legal — its kind is determined by classifying the computed multivector:

kind	rendered as
finite point	dot
ideal point	ring on horizon ellipse
line	infinite line
ideal line	dashed horizon ellipse
vector	arrow
bivector	oriented loop, area $\propto \sqrt{ b }$
rotor	arc + angle label
list	polygon outline (if all points)

Stroke widths and radii scale with the multivector's weight, so $5 \cdot e_{12}$ renders *five times as thick* as e_{12} .

The architectural contribution

Pluggable algebra adapters

Each algebra is a single file exposing one spec object:

spec field	role
Algebra, arraySize	ganja.js binding
bladeIndex, parseBladeName	basis & parser hook
classifyMV	value $\rightarrow$ kind
normalizeMVFinit/Ideal	finite & ideal norms
getRenderPlan	value $\rightarrow$ SVG primitive
supportedNodeTypes	gates parser branches
INITIAL_ITEMS	showcase on load

The parser, evaluator, expression engine and node-type registry are **factory functions** bound to a spec — **adding a new GA is one new file.**

Already shipping: PGA(2 0 1) VGA(2 0 0) R(0 1 0)

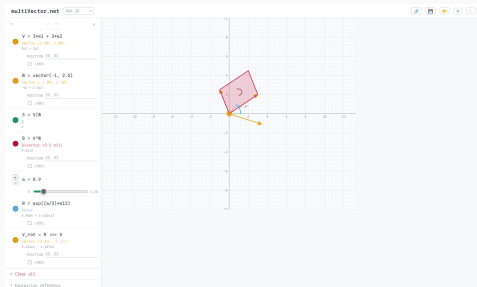
Two algebras today, more tomorrow

VGA(2,0,0) — vectorial 2D

Same expression language, same UI. Vectors, dot, wedge (bivector), exponential (rotor), sandwich.

Adding this algebra was *one file* (src/algebras/vga200/) — no changes to parser, evaluator, panel, canvas.

The same recipe will host CGA, 3D PGA, STA — limited only by what ganja.js supports.



Ganja delegation, single sign convention

Every GA primitive routes through ganja's built-in:

operation	routed to
dual !A	Algebra.Dual
reverse ~A	Algebra.Reverse
sandwich M »> A	Algebra.sw
exponential exp(V)	V.Exp() (typed copy)
sqrt(motor)	log → halve → exp
finite norm	Algebra.Length
ideal norm	Algebra.Length(Dual)

No parallel implementations to drift. Adding a new algebra inherits all of GA semantics for free — provided ganja supports it.

Features (today)

Lists as first-class values

```
L = [P, Q, M >>> P, M*M >>> P]    # any types mix
~L                                   # broadcast over list
M >>> L                             # transform every element
L1 ^ L2                             # elementwise wedge
len(L), L[i], L[i:j]                # length, index, slice
```

All unary / binary operators broadcast or work elementwise. All-point lists draw a dashed polygon.

Smart norms, postfix accessors

P & Q		# smart norm (auto finite/ideal)
A. norm	A. inorm	# explicit finite / ideal norm
abs (s)		# scalar absolute value
A.e12		# blade coefficient extraction
A.e21		# permuted blades: $A.e21 = -A.e12$

Plus norm / inorm toggle buttons per row that normalise the underlying value and propagate to dependents.

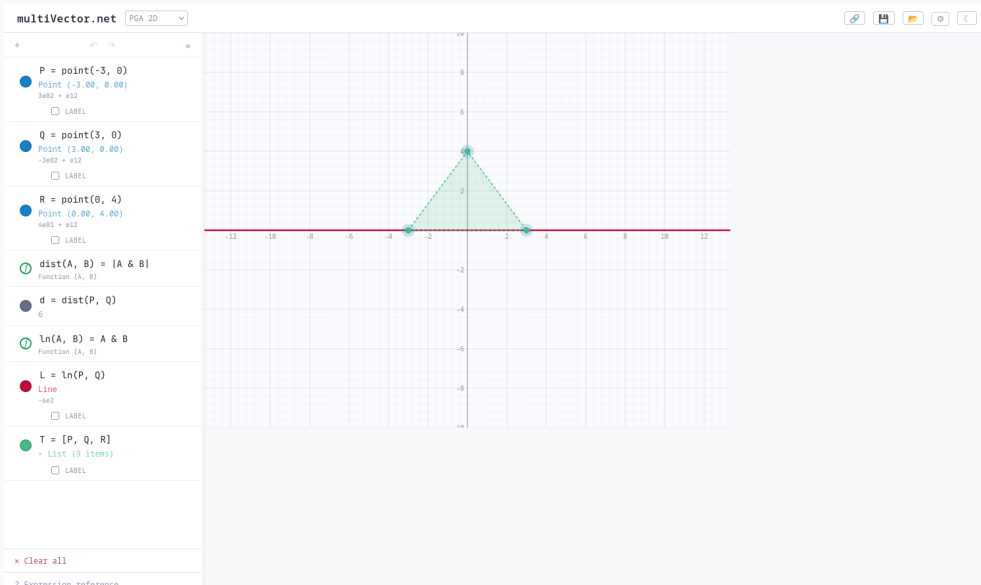
User-defined functions (just landed)

```
dist(A, B) = |A & B|           # function definition
midpoint(A, B) = (A + B) / 2
ln(A, B)    = A & B             # parameters are any object

d  = dist(P, Q)                 # call anywhere
m  = midpoint(P, Q)
L  = ln(P, Q)
d2 = 2 * dist(P, Q) + 1         # nested in arithmetic
```

Functions are values — captured globals propagate reactively, recursion is allowed with a depth cap, builtin-name collisions are rejected at parse time.

User functions, live



Scalars are sliders

- interval (min, max, step) per scalar,
- play ▷ / pause; modes: pingpong, repeat, once, infinite,
- speed levels 0.25× to 8×,
- any dependent object animates in lock-step.

Direct manipulation

- drag points, vector tail / tip independently,
- lock per item, hide per item,
- undo / redo (Ctrl-Z / Y), 100-step history with high-frequency drag coalescing,
- appearance per item: color, opacity, size, shape, label.

Polish that matters for adoption

- **Light / dark theme** — every SVG color goes through CSS variables; theme switch cross-fades.
- **Adaptive grid** — major + minor (5-subdivision) gridlines, axis labels reposition when off-screen.
- **Persistence** — autosave to `localStorage` keyed per algebra; Save / Open named graphs.
- **Share** — encode the whole graph in a URL fragment for a one-click link.
- **Help modal** — full expression reference inline, no docs round-trip needed.
- **Error UX** — duplicate labels, invalid syntax, builtin collisions all surface as a red pastille + a single-line message; the row's value display is suppressed cleanly.

Why this matters for the PhD

A platform, not a demo

Most GA visualisation work is one-off: a fixed algebra, a fixed scene, a paper figure.

multiVector.net is general infrastructure.

- Teaching: live derivations in a lecture, students fork the URL.
- Research: prototype a construction in minutes, save the JSON alongside the paper, drop a URL into reviewer correspondence.
- Community: a shared substrate where each algebra adapter and each saved graph compounds.

Concrete academic contributions

1. **The algebra-adapter pattern.** A reproducible recipe for hosting any Clifford algebra in a single interactive UI.
2. **The expression language.** A small but complete surface — binary GA operators, lists, smart norms, postfix accessors, label templates, user functions.
3. **A reference implementation.** Open, web-native, no install — every figure in a future paper can be a live URL.
4. **A library of saved constructions.** The JSON format is stable; each saved graph is a citable artefact.

Each of these is a paper-sized contribution. Together they are the beginning of a community standard.

- More algebras: CGA, STA, 3D PGA — each is one adapter file.
- Per-graph published URL → versioned share links.
- Inline LaTeX rendering of expressions and labels.
- Replay of construction history as an animation (already have the undo / redo log — just expose it).
- Export: SVG, animated GIF, PDF figure with attached `.json`.
- A gallery: curated saved graphs as a discoverable corpus.

The ask

I would like to position the publication of **multiVector.net** — the system, the language, and the adapter pattern — as a contribution of my thesis.

Short-term targets I can argue for:

- a tool paper (e.g. AGACSE, CGI tools track, or JOSS),
- a tutorial paper using the platform for derivations,
- a community-standard proposal (saved-graph JSON format).

Live demo: <http://localhost:5174>

Thank you, Vincent.

Questions, sharpened criticism, and re-direction welcome.